

Database notes – LCA Commons

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For database pack from openLCA Nexus, <https://nexus.openlca.org/>

General notes (for any data pack from Nexus): This is a zolca file which is basically a compressed openLCA database. To use it in openLCA, either **import it as a new database** (right-click in openLCA navigation panel → New database → From file; main menu function: Database → New database → From file); this will recreate the database in openLCA, as a new database. **Or, import it into an existing database** which is already open in openLCA (then right-click on the open database → import → import entire database). This allows you to combine different databases and Nexus data packs.

Specific notes for this database: The LCA Commons is a database providing US representative LCA data. The >10,000 datasets have been developed by the different US governmental agencies such as USDA, National Renewable Energy Laboratory (NREL), National Agricultural Library (NAL) and US Forest service and have been created with varying modeling perspectives and nomenclature frameworks. The database is available for free on the LCA Collaboration Server (<https://www.lcacommons.gov/lca-collaboration/search>). However, it is the user's responsibility to combine them and handle errors that exist across the repositories. The database described here was created by GreenDelta GmbH based on the public repositories, with the idea to better align the datasets with each other and to create one integrated and homogeneous database.

Creating product systems:

The setting “auto-link processes” should be used with caution when creating a product system due to:

- Presence of product and waste flows involved in waste treatments, recycling and energy recovery;
- When calculating the impact assessment, pay attention to the selected allocation method. An allocation method should be set (e.g., physical), especially in case of presence of multioutput processes like “Petroleum refining, at refinery” and “Crude oil, in refinery”.

Memory allocation: This database should not cause problems whilst calculating. Still, you can allocate more memory following File → Settings → Configuration → Increase the maximum memory usage in MB.